Solar Regenerated Daily Cycle Passive Dehumidifying Air-Conditioner: Storing Drying Potential in Solid Desiccant Through Daily Weather Cycles

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Abstract

Air conditioning is a major energy consumer, particularly in humid regions due to the high latent heat factor. (Boukhanouf *et al.*, 2015) Studies in tropical climates indicate that air-conditioning is commonly used in bedrooms during sleeping hours when ambient humidity is elevated. (Kubota *et al.*, 2010)

This paper examines the feasibility of a solar-driven desiccant dehumidification air-conditioning system that undergoes daily adsorption and desorption cycles to provide dehumidification for occupant comfort. The system utilizes free solar energy and a fan to force air through a delayed passive dehumidification process. During daytime hours, solar energy regenerates a charcoal desiccant material, storing its drying potential for later use. At night, ambient air is passed through the dry desiccant chamber, dehumidifying the air.

An experiment conducted in Sendai, Japan demonstrated this principle under real-world weather conditions. The results show a successful 40% reduction in relative humidity over a 12-hour period at night during high humidity weather, showcasing the potential for solar-powered dehumidification to supplement air conditioning.

Optimizing the system's performance involves considerations such as ventilation rate, solar energy capture efficiency, desiccant mass-to-surface area ratio, and the implementation of weather sensors to better preserve drying potential during rainy periods.

Keywords: Passive Techniques, Air Conditioning, Desiccant Humidity Control, Solar Regeneration, Daily Cycle

1. Introduction

The International Energy Agency forecasts a significant expansion in the global stock of air conditioners in buildings, from 1.6 billion currently to 5.6 billion by 2050, driven by factors such as population growth, rising incomes, falling equipment prices, and urbanization. This growth is particularly pronounced in the developing world, where usage levels currently lag behind more developed nations. (Bryant, 2019)

Studies conducted in Southeast Asia reveal that homeowners with air conditioning units typically install them in bedrooms and operate them predominantly at night during sleeping hours. (Kubota *et al.*, 2010) These regions face a substantial cooling load due to the high latent heat in the ambient air at night. Conventional air conditioning systems typically cool the air below the dew point to remove moisture, which is an energy-intensive process.(Boukhanouf *et al.*, 2015)

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Observations indicate that the outdoor ambient temperature in a tropical city can reach as high as 36°C with 40% relative humidity during the day, while nighttime conditions can drop to as low as 24°C with 90% relative humidity. (Kubota, Chyee and Ahmad, 2009)

This suggests that the primary task for residential air conditioners in these regions is to dehumidify the air at night to achieve a comfort level of approximately 25°C and relative humidity below 65%. (ASHRAE, 2021)

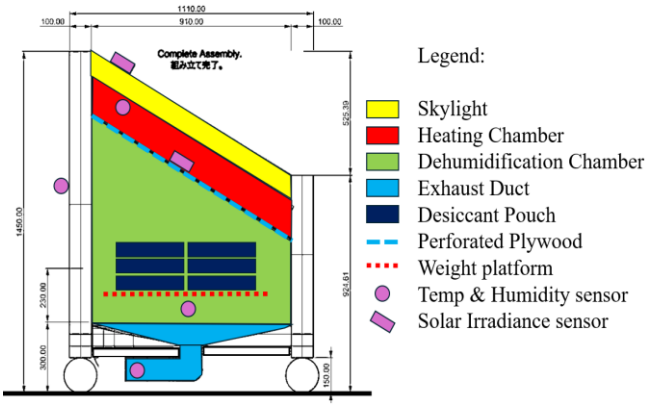

Fig. 1 Experiment Box Sectional Diagram

1.2. Research Question

This study investigates the feasibility of a passive or semi-passive dehumidification approach that leverages the natural diurnal fluctuations in humidity. Specifically, it hypothesizes that the dry daytime air can be utilized to regenerate desiccant materials, which can then be employed for dehumidification during the humid nighttime hours.

Accordingly, this research aims to address the following research questions:

a. What level of passive air dehumidification can be achieved by leveraging the diurnal variations in humidity?

b. How effectively can desiccant materials absorb nocturnal humidity and then be regenerated using solar energy as the heat source?

c. To what extent can a desiccant material store dehumidification potential for later use when ambient humidity levels rise?

d. What are the climatic factors that need to be considered to potentially automate a passive humidity control system based on the natural humidity cycles?

2. Methods

2.1. Experiment System Design

The proposed system utilizes a solar-powered desiccant dehumidification approach, which features a substantial delay between the regeneration and dehumidification phases. The design aims to remove nighttime latent heat by passing outdoor air through a solid desiccant bed. The desiccant is then regenerated in the daytime, enabling repeated cycles of dehumidification and regeneration.

The physical experiment was conducted in Sendai, Japan, from July 2023 to June 2024 to assess the system's feasibility under real-world weather conditions. The location was selected to facilitate close monitoring of the experimental setup and apparatus. Recorded Data indicates that the monthly mean daily maximum temperature in Sendai during July and August 2022 was 28.7°C and 29.4°C, respectively, with monthly mean relative humidity of 85% and 81% respectively ('Japan Meteorological Agency', 2023). It was believed that the hot and humid summer conditions in Sendai, Japan could provide a reasonable comparison for the weather in other hot and humid regions relevant to the study.

Multiple experimental cycles were carried out, with the amount of desiccant and ventilation speed rates as the manipulated variables between cycles. The experiment was later repeated during the winter months to evaluate the system's performance under colder conditions, as an additional finding for its potential applications.

2.2. Experiment Components

The experimental apparatus is constructed within a wooden frame box measuring 1m wide, 1m deep, and 1.5m tall as seen in

Fig. 1. The enclosure is fabricated from 6mm plywood panels with 50mm thick extruded polystyrene insulation lining the interior to minimize heat loss from the system. A skylight comprised of two 3mm thick twinwall polycarbonate panels with a 25mm air gap in between is installed at a 30-degree angle to the horizontal plane, facing the south. Underneath the skylight, a black perforated plywood panel serves to capture solar radiation.

Outdoor ambient air is drawn into the space between the skylight window and the perforated black plywood panel using a ventilation fan where it is heated by solar irradiance in the daytime. The plywood panel is perforated to allow air to pass through to the dehumidification chamber below.

Within the dehumidification chamber, air comes into contact with the desiccant material. Desiccants are placed in air-permeable bags atop a metal mesh. Here the desiccants interact with the air flow before being exhausted out from below.

Air is exhausted from the experimental enclosure by a timer-controlled ventilation fan equipped with one-way dampers. The fan operates from 8 AM to 4 PM during the solar irradiance period for desiccant regeneration, and from 6 PM to 6 AM during the nighttime period of increased relative humidity for the air dehumidification process.

2.3. Desiccant used

Charcoal flakes derived from recycled wood was utilized as solid desiccant owing to it being a sustainable and readily available material. The desiccants were distributed in air-permeable pouches into 1.5kg bags as seen in Fig. 15. Multiple bags were arranged on a metal mesh tray for constant weight measurement. Various quantities of bags were assessed throughout the experimental trials, ranging from 2 bags to the maximum of 6 bags.

2.4. Measured data

Fig. 2 shows the multiple sensors that were deployed throughout the experiment to monitor the system's performance and behavior. Data was collected at 2-

Fig. 2 Table of Sensors.

Sensor Type	Position	Sensor model	Measured	Converted
Temp & Rel. Humidity	2 Exterior & 5 Interior	T and D TR72Ui	°C, %	g/kg
Solar Radiation	exterior & Top Interior	EKO ML-01	µV/Wm²	W/m²
Ventilation Rate	Intake & Exhaust Duct	Testo 400Vane Probe	m/s	m³/hr
Weight scale	Under Desiccant tray	AND GP-30K	gram	Kg

minute intervals to ensure high-resolution data collection. All the Data was recorded on a connected windows computer.

Temperature and humidity measurements were obtained from various locations within the experimental setup, including the outdoor ambient air, the entrance and top of the heating chamber, the top of the dehumidification chamber, below the desiccant materials, the exit of the dehumidification chamber, and the end of the exhaust duct. These sensors were strategically shielded from direct solar radiation to ensure the accuracy of the air data collected.

2.5. Calculated Data

Absolute humidity values from collected temperature and relative humidity data were calculated using the Hyland and Wexler equation as shown below.

$$P_s = (\exp(-\frac{5800.2206}{t+273.15} + 1.3914993 - 0.048640239 \times (t+273.15) \times 10^{-4} \times (t+273.15)^2 - 0.14452093 \times 10^{-7} \times (t+273.15)^3 + 6.5459673 \times \log(t+273.15)))/1000$$

$$P = P_s \times \frac{RH}{100}$$

$$x = 0.622 \times \frac{P}{101.1325 - P} \times 1000$$

Where, P_s [kPa] is saturated water vapor pressure, t[°C] is temperature, P[kPa] is water vapor pressure, RH[%] is relative humidity and x [kg/kg'] is absolute humidity.

3. Results

3.1. Solar Irradiance

Solar irradiance data was collected by placing the sensor on the inclined plane of the enclosure next to the skylight. An additional sensor was placed below the skylight, on the perforated plywood panel to quantify solar energy entering the enclosure. From Fig. 3 we see the solar irradiance peaking at approximately 1200 W/m^2 on clear, sunny days while the maximum efficiency of solar collector was around 80%.

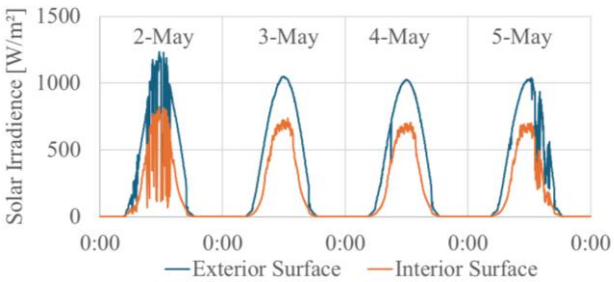

Fig. 3 Solar irradiance data over clear sunny days.

3.2. Air Temperature

In Fig. 4 during peak solar irradiance, we see the air temperature within the heating chamber increased by 70°C compared to the ambient outdoor air. The Air temperature in the Dehumidification chamber is reduced by approximately half. The final exhaust air temperature exceeded the outdoor air temperature by only 3°C.

At night, the temperature in the dehumidification chamber remained relatively elevated. The exhaust air was warmer than the outdoor air by about 2°C.

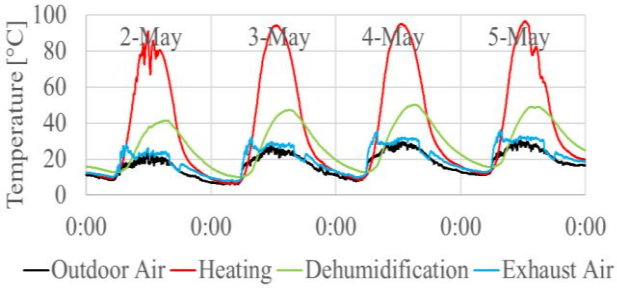

Fig. 4 Temperature at various points in the system over high solar irradiance.

3.3. Relative Humidity

The relative humidity of the exhaust sensor rises and falls in the early morning and late evening hours corresponding to the activation and deactivation of the ventilation fan shown in Fig. 5. During these deactivation periods, the exhaust sensor detects the outdoor ambient air conditions.

In the regeneration phase, the relative humidity within the heating chamber can drop to nearly zero due to the high air temperatures. During the dehumidification phase, the relative humidity of the exhaust air is approximately 40%. This is significantly lower than the 80% relative humidity of the outdoor air.

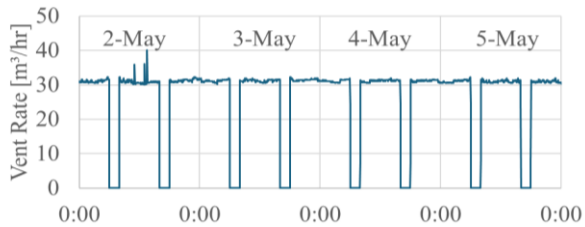

Fig. 5 Ventilation rate and Fan activation periods.

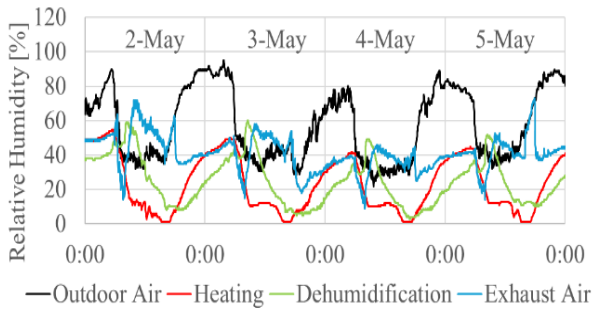

Fig. 6 Relative Humidity over high solar irradiance.

3.4. Absolute Humidity

It can be seen in Fig. 7, during the regeneration phase, the absolute humidity value of the exhaust air can double that of the outdoor ambient air. Conversely, during the nighttime dehumidification stage, the exhaust air absolute humidity is reduced to by half of the outdoor air.

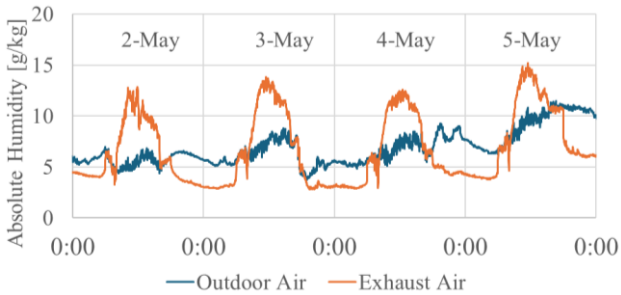

Fig. 7 Absolute Humidity over high solar irradiance days.

3.5. Desiccant Weight

As the desiccant material absorbs moisture from the air, a corresponding increase in its weight can be observed. The rate at which the desiccant can be regenerated is higher than the rate of adsorption. In Fig. 8, six bags of 1.5 kg desiccants, totaling 9 kg, facilitated the adsorption of approximately 0.15 kg of water over the dehumidification cycle.

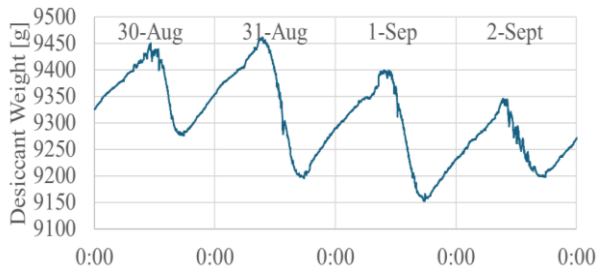

Fig. 8 Desiccant weight Data over high solar irradiance days.

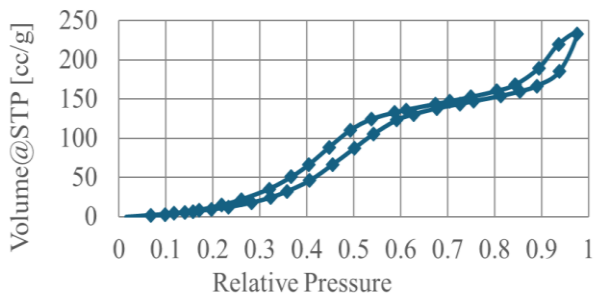

Fig. 9 Adsorption Isotherm of 0.03g Desiccant.

Although the desiccant adsorbs a reasonable amount of moisture from the air, this adsorbed value is much lower than the adsorption isotherm results obtained by performing adsorption and desorption tests on a small 0.03g sample of the same desiccant under constant 25°C temperatures and varying relative humidity conditions shown in

Fig. 9. This shows that the desiccants capacity has not been fully utilized and there is room to improve the adsorption efficiency.

4. Analysis

4.1. Ventilation Rate

The experiment manipulated ventilation rates as a controlled variable. The effects of various ventilation rates were averaged across multiple days in Fig. 10.

During the day, ventilation rates of 42 m³/hr and 48 m³/hr facilitated better regeneration compared to 61 m³/hr. The slower rates allowed more heat buildup, which enabled greater moisture release from the desiccant. However, further reducing the ventilation rate deteriorated the system's performance. The system could not expel enough moisture from within the enclosure at very low ventilation rates.

In contrast, slower ventilation rates down to 21 m³/hr performed better for nighttime dehumidification. Yet, the slowest rate of 6 m³/hr did not perform as well. The extra contact time between the air and the desiccant allowed for improved adsorption.

The optimal ventilation rates for the regeneration period differed from those that worked best during the dehumidification period.

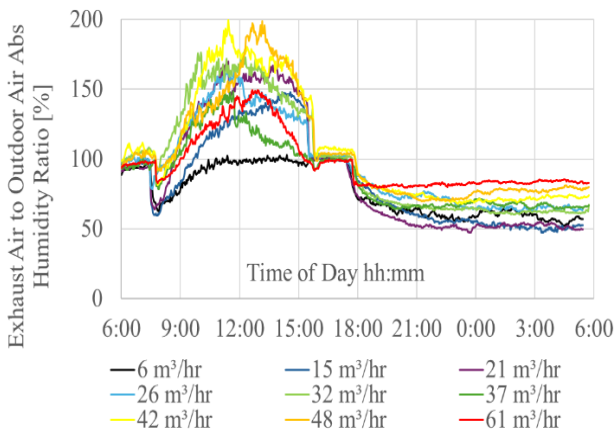

Fig. 10 Humidification and dehumidification effect of various ventilation rate setting.

4.2. Desiccant Capacity

Fig. 11 shows the dehumidification ratio of processed exhaust air absolute humidity and outdoor air absolute humidity under the varied desiccant quantities used in separate experiment cycles. The analysis shows that greater desiccant volumes led to slightly enhanced adsorption during the dehumidification phase. However, the benefit of increasing the desiccant amount diminishes as more desiccant is added. This

indicates that just increasing the amount of desiccant is ineffective unless the contact efficiency is improved.

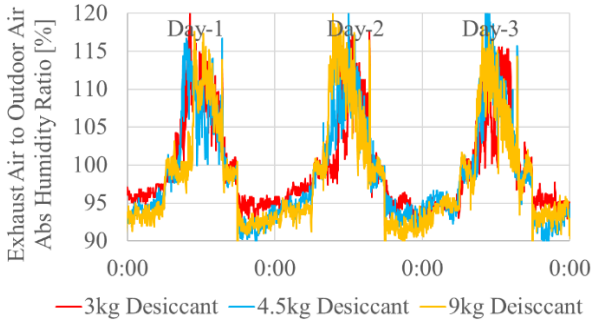

Fig. 11 Dehumidification effect of varying amount of desiccant used at constant ventilation rates.

4.3. Drying potential

It has been observed that improved desiccant drying during the day enhances the material's capacity to absorb moisture at night. By utilizing a lower ventilation rate, the additional solar radiation per unit air volume generates higher temperature air. Consequently, the desiccant releases more humidity, resulting in greater drying potential that enhances the adsorption performance in the dehumidification phase.

Fig. 12 compares two different ventilation rate settings during the regeneration period. The first setting used a ventilation rate of 35 m³/hr, leading to less desorption, while the second setting used a ventilation rate of 26 m³/hr, resulting in better desorption. The nighttime ventilation rate was the same for both at 26 m³/hr. The desiccant that underwent better regeneration exhibited a higher drying potential in the dehumidification phase.

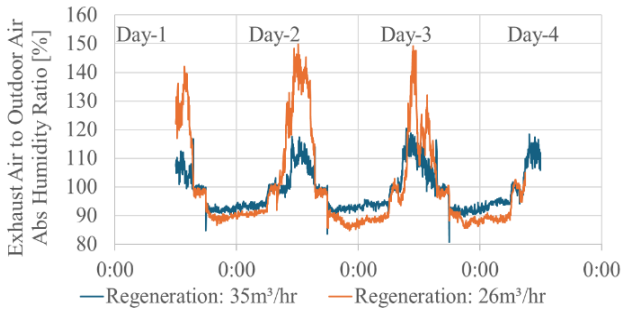

Fig. 12 Dehumidification effect of varying ventilation rates during regeneration.

5. Discussion

5.1. Desiccant form

The desiccant's physical form is predicted to significantly impact the adsorption and desorption rates, as well as the drying potential capacity of the desiccant.

The desiccant's adsorption and desorption processes can only occur at its external surface. The release rate from the surface to the air, and from the desiccant's internal mass to its surface, may differ, potentially introducing resistance to the adsorption-desorption rate.

While the surface area can enhance the desorption and adsorption rate, the internal volume determines the total drying potential stored in the desiccant.

An optimized desiccant form factor is predicted to contribute to the desired performance effectiveness of the overall system. (Fig. 13)

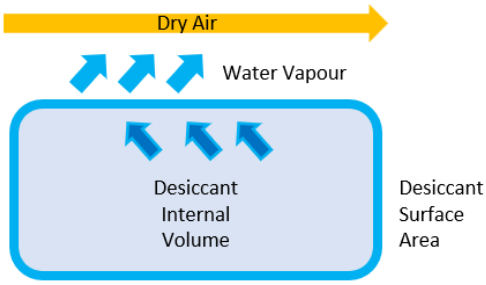

Fig. 13 Illustration of Surface area and Internal massing effect on Adsorption-Desorption rates.

5.2. Weather Awareness

The timer-controlled ventilation system used in the experiment attempts to regenerate the desiccant material even on rainy days, which conversely leads to an increase in humidity within the system instead of regeneration as seen on 6 and 8 September in Fig. 14. This ultimately diminishes the system's dehumidification capacity and performance when it is needed.

An enhanced smart sensor-based system could be programmed to monitor the external weather conditions and intelligently deactivate the ventilation on high humidity days, thereby preserving the drying potential of the desiccant material within.

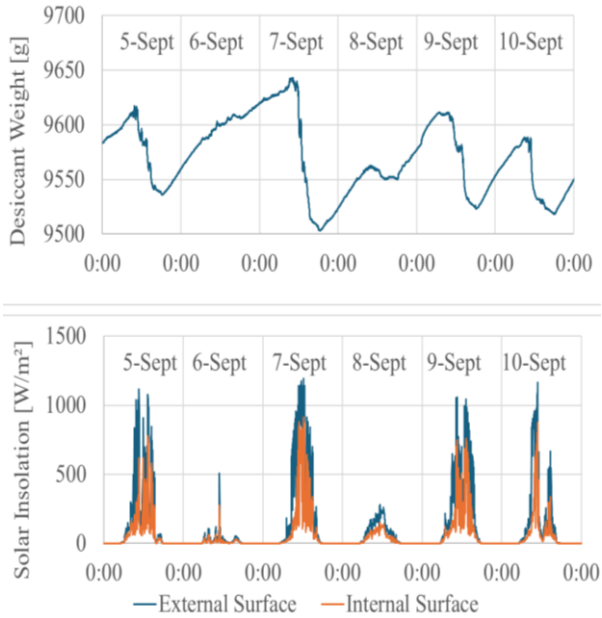

Fig. 14 Desiccant adsorption weight data over unstable solar irradiance.

5.3. Thermal Barrier

Solid desiccants have the potential to serve as thermal buffers for buildings. The solar energy absorbed by the desiccant is initially utilized to change liquid water into water vapor, which may help mitigate sensible heat gain into the occupied spaces. As demonstrated in Fig. 4, the air temperature experiences a significant decrease upon regenerating the desiccant. This hints at its use as a regenerative heat shield as a temperature Phase change material.

5.4. Winter Potential

Evaluations of the system during the winter months yielded comparable outcomes. This suggests that the system can be leveraged to extract moisture from the nighttime atmosphere and to subsequently release it under solar irradiance. This process can be harnessed to provide passive humidification through the drier winter months.

Even though the absolute humidity levels are lower during the winter season compared to summer, the system's performance remains substantial when assessed in terms of relative humidity.

5.5. Computer simulation

Developing the system and refining the component designs would benefit from a computer simulation approach. The existing physical experimental method takes multiple days for data collection and is vulnerable to volatile weather conditions. This makes it almost impossible to replicate multiple specific scenarios for direct data comparison. A simulated environment would enable the assessment of diverse factors influencing performance, leading to consistent and reliable outcomes.

It is necessary to evaluate different scenarios to comprehend and refine the system's performance. This involves exploring various system designs, simulating the long-term energy cost advantages, and assessing the systems' implementation in diverse geographical locations across the globe.

5.6. Climatic variations

The author notes that the system is conceptualized based on the air conditioning usage patterns of hot and humid regions. Although this research was conducted in Sendai, the results obtained here show that it may be applicable to tropical regions as well.

The system configuration may necessitate some fine-tuning in different geographic areas. This is because the system's operation is contingent upon solar radiation, weather conditions, and the air conditioning needs of each location. For instance, hot-humid or cold-humid regions may require sufficient drying capacity, whereas dry regions can utilize a desiccant that spreads the releases moisture throughout the day.

Further research is needed to investigate the system's performance in diverse climatic conditions to enhance its real-world applicability.

6. Conclusions

The system presents promising prospects for reducing air-conditioning demands during both summer and winter by harnessing passive renewable energies that tap into the daily humidity fluctuations in the surrounding air. Presently, the research remains at an early stage, necessitating further refinement to fully unlock the potential of the phenomena and to utilize it for passive comfort air conditioning.

The passive processes and natural energy source of the system render it highly dependent on the Earth's natural rhythms, suggesting that a method to synchronize the system design with the Earth's rhythm could enhance the system performance.

Moving forward, constructing a full-scale prototype for household cooling tests would be advantageous to gather application-specific data.

7. Acknowledgement

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Fig. 15 Desiccant Pouch in dehumidification chamber.

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